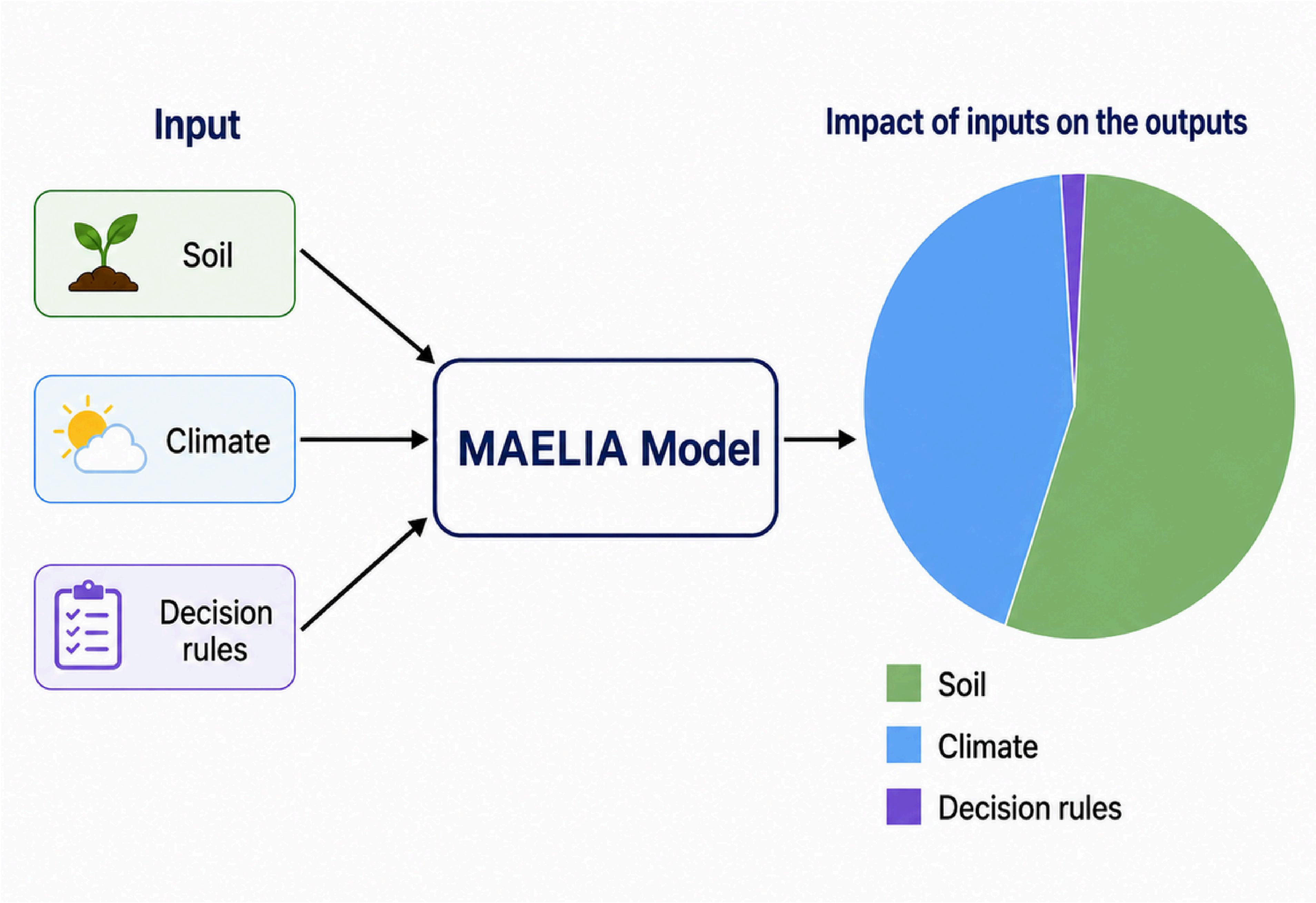
Sensitivity Analysis for MAELIA

Project logic



Key Idea

This project uses MAELIA simulations to explore how agricultural management choices can influence model outputs.

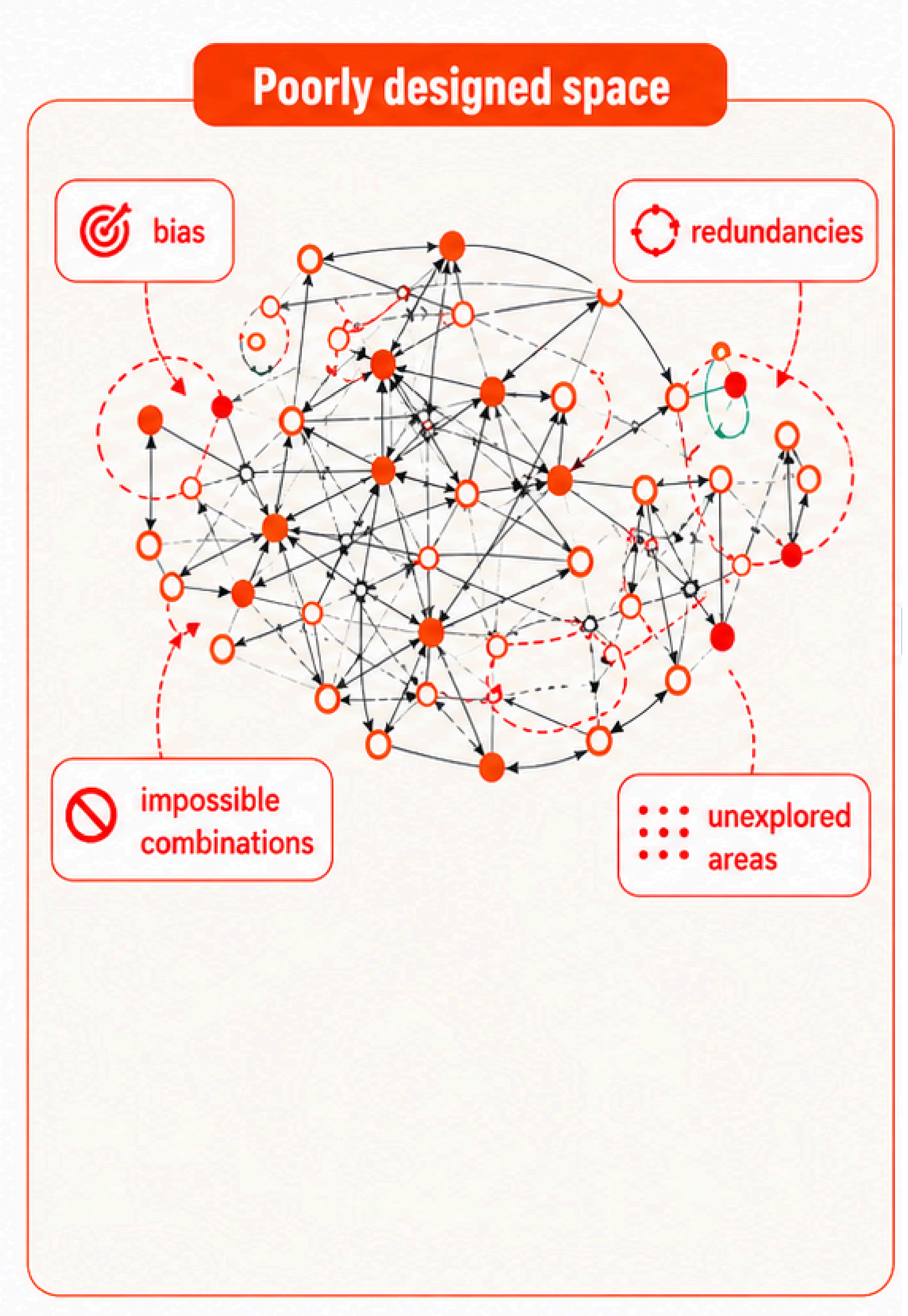
terrainSA in this project

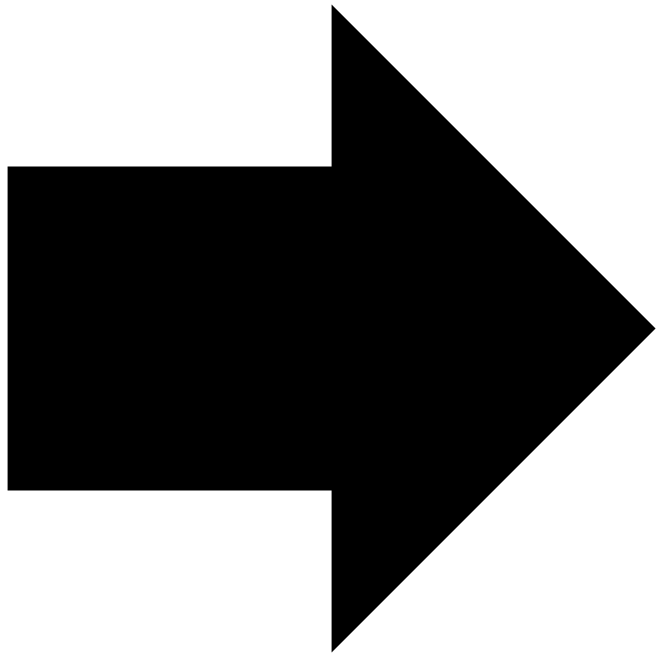
terrainSA removes soil and climate variability by iterating on the same plot. This plot was validated by agronomist Olivier Therond.

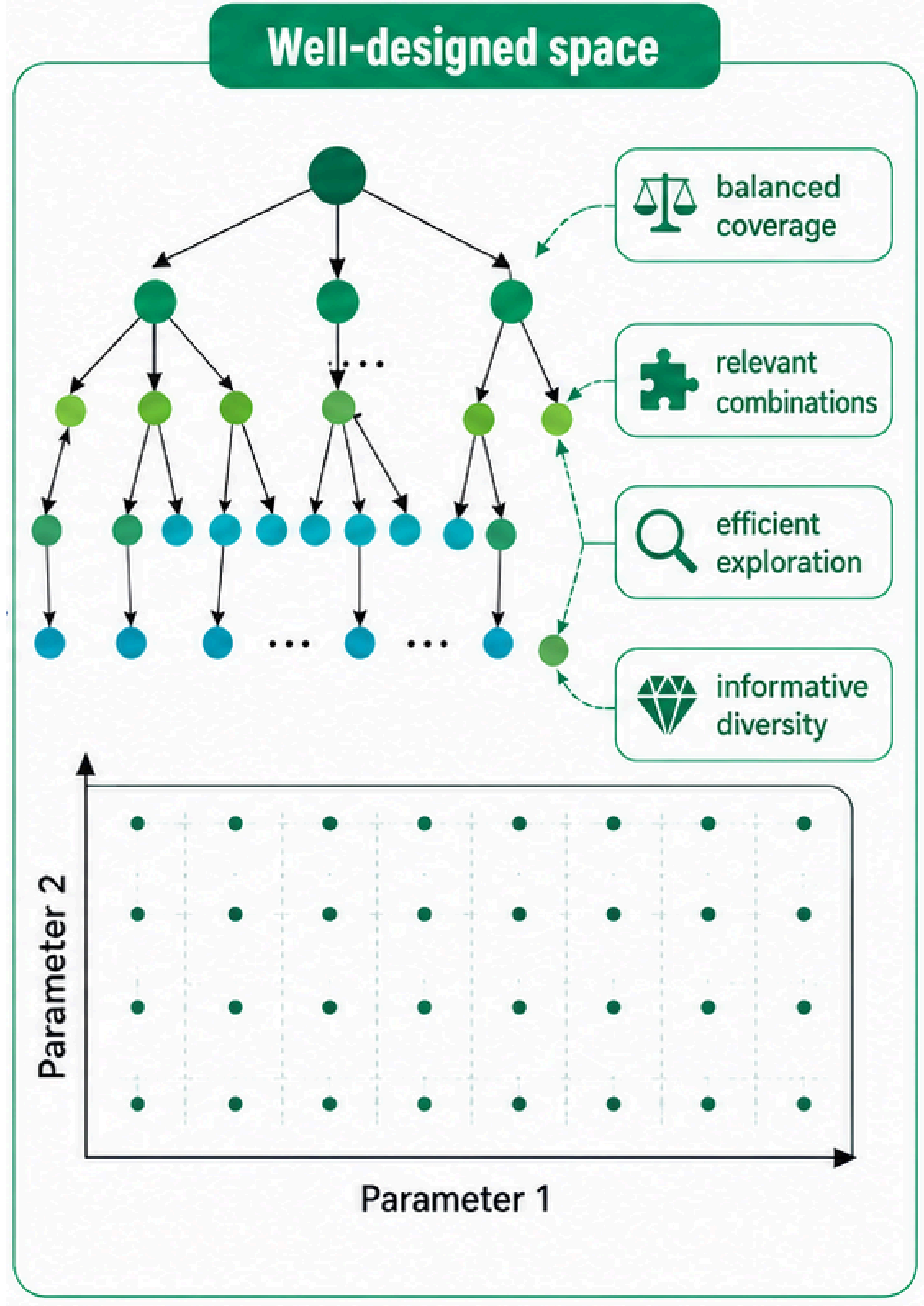
Why control the context?

Soil and climate can strongly shape simulated behaviour. To study technical operations more clearly, the analysis needs a setting where those effects are not mixed together.

Designing the parametter space







Step 1

Setting up the sampling strategy to efficiently explore the parameter space.
(In collaboration with Paul Saves)

Step 2

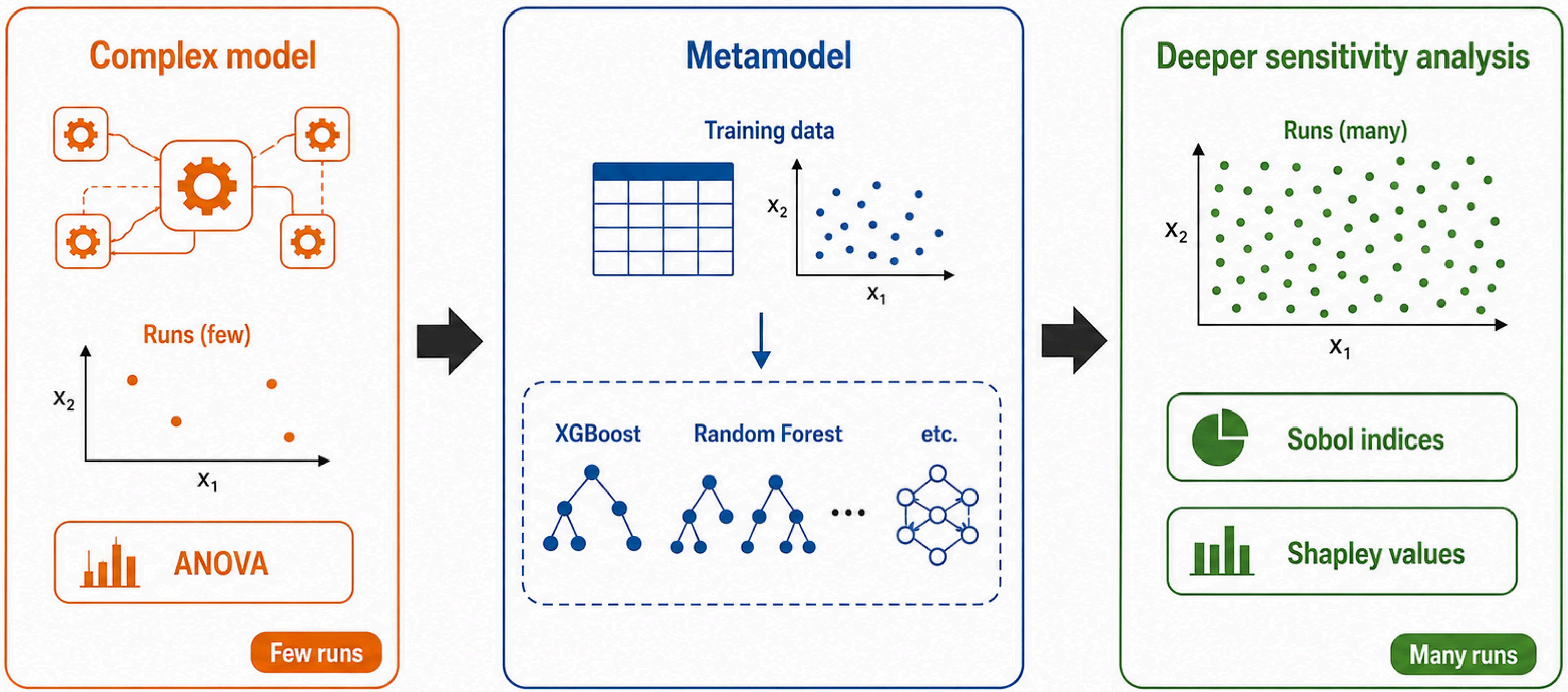
Iterating on the same land parcel to isolate technical operations using terrainSA.

Step 3

Using a Python script to analyze simulation results through ANOVA, Sobol indices, Shapley values, decision trees, and other statistical methods.

Step 4

Interpreting the results with the team to improve the sensitivity analysis, define next steps, identify methodological improvements, and refine regions of interest.



The role of metamodels

WebApp demonstration http://127.0.0.1:8000/

Thank you!

Any questions?